

Sandstorm_Mod_Manager
Created for Insurgency Sandstorm 1.21 by Joanna Wick
Version: 1.2.1
Date: 2026-07-25

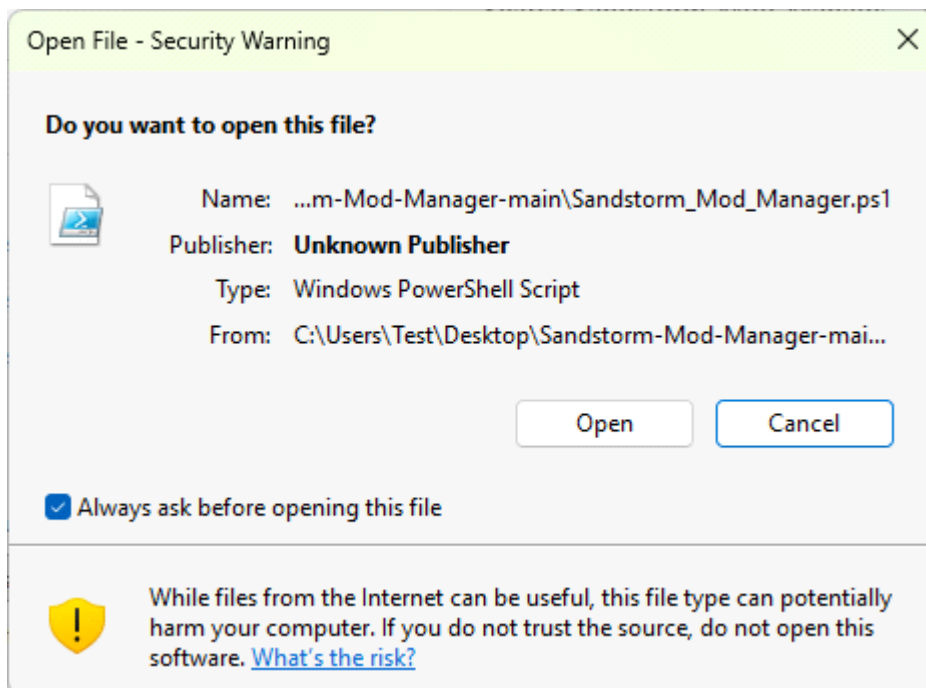
What is new

1.2.5 (2026-08-05)

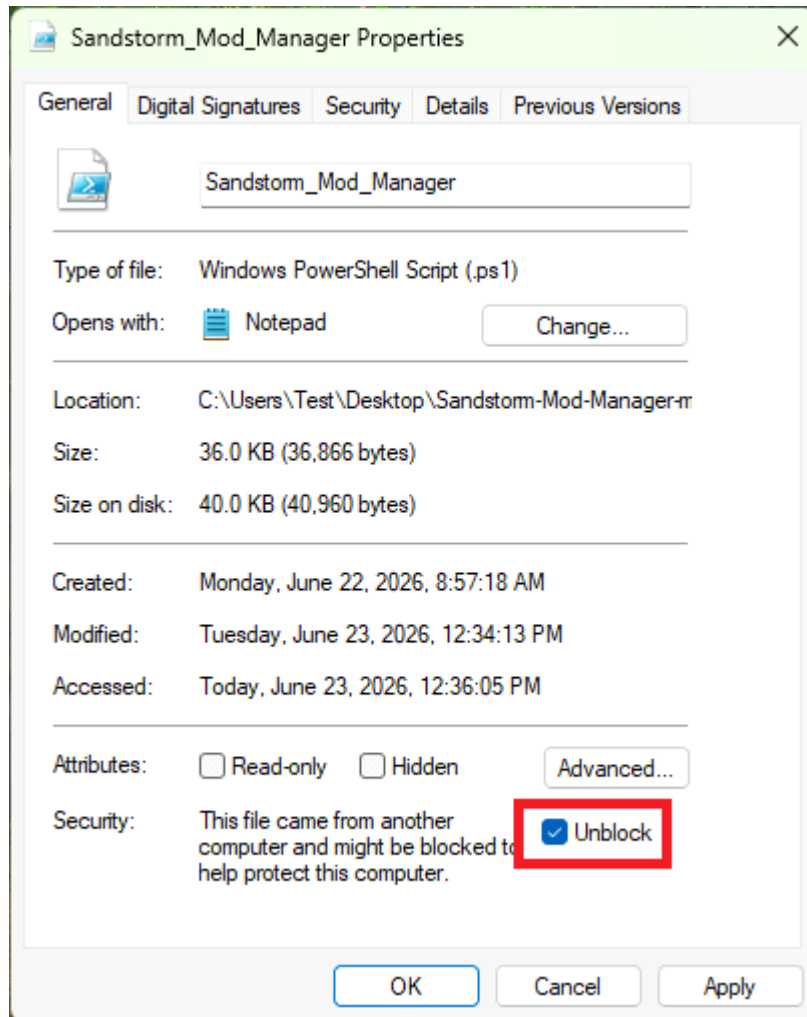
1. [ADDED] On first execution script will scan state.json to build ModList.json. This will prevent downloading of all subscribed mods the first run.
2. [ADDED] "Select All" and "Clear All" buttons to the Force Download and Un-Subscribe GUIs. This way you can force a re-download of all mods or a mass unsubscribe of all mods without having to go to Mod.io
3. [ADDED] "Total Subsriptions: " to Main Menu
4. [ADDED] "Mod Storage: " showing total disk storage for all Mod downloads to Main
5. [ADDED] Directory path to where mods stored shown below Move Mod.io selection.

*** NOTICE * * NOTICE * * NOTICE * * NOTICE ***

Double-Click on Start_Sandstorm_Mod_Manager.bat to start the script. Do not double-click on Sandstorm_Mod_Manager.ps1



You will see the above window open when you try to run the Start_Sandstorm_Mod_Manager.bat. To stop this window from being opened Right Click on Sandstorm_Mod_Manager.bat. and click on Properties.



Check the highlighted Unblock and click Apply. You should no longer see that window popup.

Introduction

=====

This PowerShell script manages all of your subscribed mods for Insurgency Sandstorm from Mod.io and install them for use with your Sandstorm game on Windows and Windows Servers. It will also update the state.json file located in the mod.io/254/metadata directory. This is where the game stores all information about each mod that it uses in the game. This script will also download, install and update the state.json file for any new mods you have subscribed without having to load the game.

If you want to make sure any new mods or new updates are installed just run this script before playing

the game. The scan is very fast. Any updated mod or map will be downloaded and installed as well as anything new you have subscribed. You won't have to worry if the game is actually downloading anything anymore when you start the game.

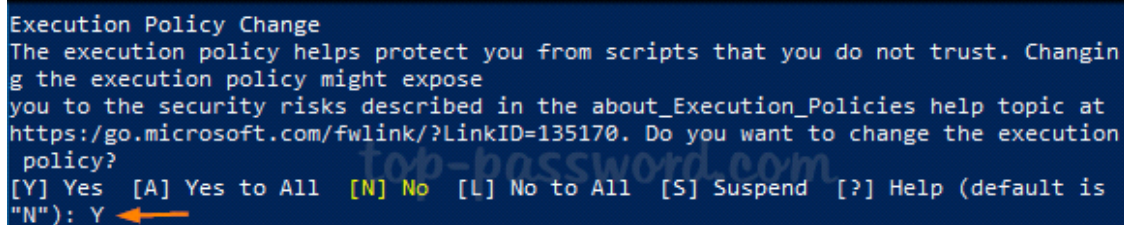
The Download Progress Bar has been disabled because it slows the actual downloads to a crawl. A 1MB download could take 15-20+ minutes no matter how fast your connection. With the Download Progress disabled the downloads are as fast as your connection speed. After downloading you will have an un-zip progress indicator that will run for each download.

The script will only download and update any files that have changed on mod.io since the last time the script was run.

If you Unsubscribe to any mod on Mod.io just run this script and it will delete the directories containing those mods so you will not need to do it manually.

First Time running Powershell Script

=====

A screenshot of a PowerShell terminal window with a dark blue background and white text. The text reads: "Execution Policy Change", "The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose you to the security risks described in the about_Execution_Policies help topic at https://go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution policy?", "[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is \"N\")". An orange arrow points to the 'Y' in the prompt. A faint watermark "top-password.com" is visible in the background.

```
Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing
the execution policy might expose
you to the security risks described in the about_Execution_Policies help topic at
https://go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution
policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is
"N"): Y
```

If you haven't run a Powershell script before or if your permissions are not defined you may see the above message then you run the script. All you need to do is enter an A or Y and the script will run without issue. If you enter N, L or S the script will NOT run and you may have a hard time getting it to execute without making policy changes. So, do not enter N, L or S.

Sandstorm Mod Manager Menu

=====

```
C:\WINDOWS\system32\cmd.exe

=====
Sandstorm Mod Manager
Version: 1.2.5
=====

Success: Valid OAuth Token!
Authenticated as: SandyD
User ID: ██████████
Total Subscriptions: 92
Mod Storage: 140.76 GB

1. Process Mod Subscriptions
2. Select Mods to FORCE Download and Update
3. Select Mods to Unsubscribe and Delete
4. Would you like to view Verbose Mod Information (not saved)
   Current Setting: Concise
5. Move Mod.io Mod Directory to new Location
   Current Mod Storage: F:\SteamLibrary\steamapps\common\sandstorm\Insurgency\mods\mod.io\
6. Exit

=====
Please enter your selection (1-6):
```

The main Menu will now show how many mods you are subscribed and how much disk space they are using. The current path for storing Sandstorm Mods is shown below the Move Mod.io Mo Directory Selection.

1. Process Mod Subscriptions

=====

```
Windows PowerShell

=====
Downloading your Subscriptions
=====

Found 1 Subscription Page.
Found 72 subscriptions.

=====

Processing 1 of 72 Subscriptions : Warehouse
modid: 6130558
modFilename: modfile_6130558_a.zip
filesize: 639.38 MB
moddate_added: 2026-06-11 22:52:42
modVersion: 1.0.1
modURL: https://g-254.modapi.io/v1/games/254/mods/6130558/files/7839678/download

Skipped Warehouse up to date
=====
```

Mod.io only allows a maximum of 100 mods to be retrieved from your subscriptions each time. They are divided into pages. You will be shown how many pages have been found and how many mods you are subscribed. The above image shows the Verbose setting to show the modid, filename, compressed size,

date added/updated, version and the direct URL used for downloading.

```
Windows PowerShell

=====
Download your Subscriptions
=====

Found 1 Subscription Page.
Found 72 subscriptions.

=====
Processing 1 of 72 Subscriptions : Warehouse
Skipped Warehouse up to date
=====
Processing 2 of 72 Subscriptions : Estates
Skipped Estates up to date
=====
Processing 3 of 72 Subscriptions : Mashtuur
Skipped Mashtuur up to date
=====
Processing 4 of 72 Subscriptions : Daybreak
Skipped Daybreak up to date
=====
Processing 5 of 72 Subscriptions : Hacienda
Skipped Hacienda up to date
=====
Processing 6 of 72 Subscriptions : Compass Radar Remake
Skipped Compass Radar Remake up to date
=====
Processing 7 of 72 Subscriptions : The Open Pit
Skipped The Open Pit up to date
=====
Processing 8 of 72 Subscriptions : NoRestrictPlus
Skipped NoRestrictPlus up to date
```

This is the regular condensed display.

```

Processing 70 of 72 Subscriptions : More Ammo Mutator [1.15]
modid: 98373
modFilename: filedata-5qlr.zip
filesize: 0.01 MB
moddate_added: 2023-10-30 19:46:51
modVersion: 0.8.4
modURL: https://g-254.modapi.io/v1/games/254/mods/98373/files/4468622/download

  Downloading More Ammo Mutator [1.15] - 98373 - filedata-5qlr.zip - 0.01 MB
  Unpacking More Ammo Mutator [1.15] - 98373 - filedata-5qlr.zip
7-Zip (a) 26.01 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-04-27
Scanning the drive for archives:
1 file, 12191 bytes (12 KiB)

Extracting archive: zip\filedata-5qlr.zip
--
Path = zip\filedata-5qlr.zip
Type = zip
Physical Size = 12191

Everything is Ok

Size:          48477
Compressed: 12191
Success: Archive extracted with no errors.
=====

```

When mods are being downloaded there isn't a progress bar shown. I found that a 1MB download would take 15-20 minutes to download no matter what your connection speed. I have disabled the progress bar. The file will unzip to the mods directory with a progress indicator after downloading is complete.

```

=====
Processing 64 of 72 Subscriptions : Winter
  Downloading Winter - 164381 - filedata-0yto.zip - 266.93 MB

  Download complete!
  Time Elapsed : 7.25 seconds
  Download Speed: 294.66 Mbps

  Unpacking Winter - 164381 - filedata-0yto.zip
7-Zip (a) 26.01 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-04-27

```

After a file has downloaded it will display the amount of time it took to download and the approximate download speed.

```

Success: Archive extracted with no errors.
=====

Total Download Time: 21:51
Total Download Size: 46.96 GB
Total Disk Storage : 99.71 GB

=====
Successfully Installed: 40 file(s)

```

When all mods have been downloaded you will be shown how long it took, total GB of files downloaded

and total Disk Space used.

```
=====
Successfully Installed: 3 file(s)
Success: Mashtuur - 5844541 - modfile_5844541.13.zip no errors.
Success: Insurgency 2014 Map Pack - 851078 - filedata-8nui.zip no errors.
Success: More Ammo Mutator [1.15] - 98373 - filedata-5qlr.zip no errors.
=====
```

A list of all successfully installed mod files is shown. Any downloads that give warnings during unzip will be listed next. The warnings can usually be ignored. Finally any downloads with errors will be listed. You should scroll up to see if they were fatal. You can always use the FORCE download option to download those files again.

NOTE: If you see a zip file had CRC-ERRORs while unpacking it means the calculated CRC doesn't match the one in the zip file. The contents of the zip file will still be unpacked and usable by Sandstorm. Sometimes 7-Zip will claim it's a Fatal Error when in fact it isn't for Sandstorm. I have tested this on a map called Daybreak (if the map is updated the error may go away). When it is being unzipped a Fatal CRC-ERROR occurs but the files are still unpacked and the game plays the map without issue.

```
=====
Checking for Unsubscribed Mods
=====
Deleting Unsubscribed Mod: 4630134 - Factory
=====
Sandstorm Mod File Update Complete
=====
Press Enter to continue...:
```

If there were any unsubscribed mods found the directories containing them will be deleted.

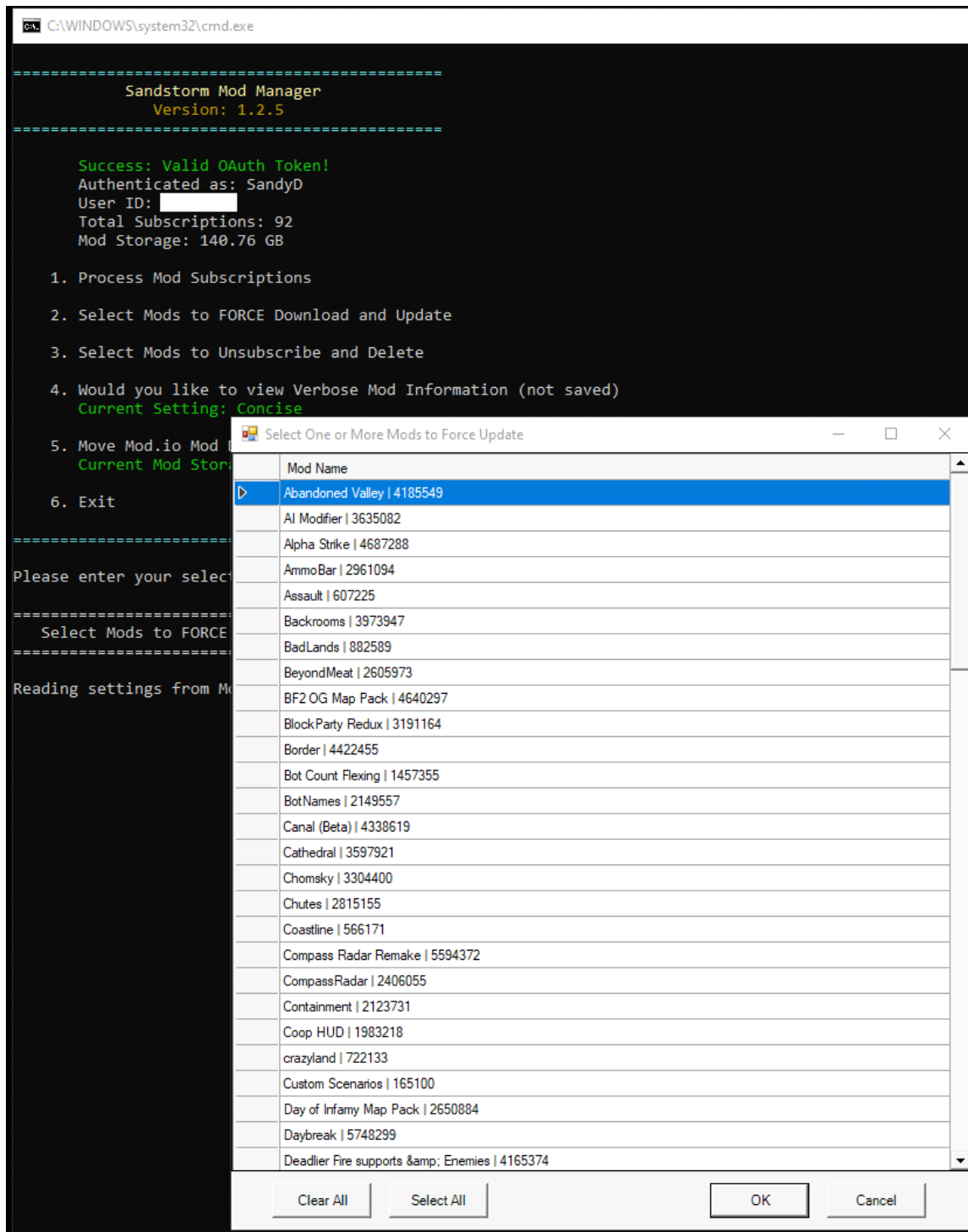
Once completed Sandstorm will have all mods updated and a new state.json file created. You can load the game and shouldn't have anything download. It could still happen if a mod was updated between the time you ran the script and loaded the game. You will still see the Downloading Server Mods... screen if the server adds new mods since the last time you joined. If on the remote chance it does stop on the Downloading server mods... screen after running this script you can exit the game to Desktop and reload and the problem should be solved. Just another one of the weird quirks with how the game incorporated the mod.io system.

Once completed Sandstorm will have all mods updated and a new state.json file created. You can load the game and shouldn't have anything download. It could still happen if a mod was updated between the time you ran the script and loaded the game. You will still see the Downloading Server Mods... screen if the server adds new mods since the last time you joined.

If you have subscribed to mods directly through the mod.io website you can run this script and update your mods to include them before you run the game.

2. Select Mods to FORCE Download and Update

=====



A GUI will open showing a list of all the mods you are subscribed. You can click on as many mods as you would like. The GUI uses the standard method for selecting multiple items on Windows. You now have the option to Select All or Clear All.

```
=====
Select Mods to FORCE Download and Update
=====

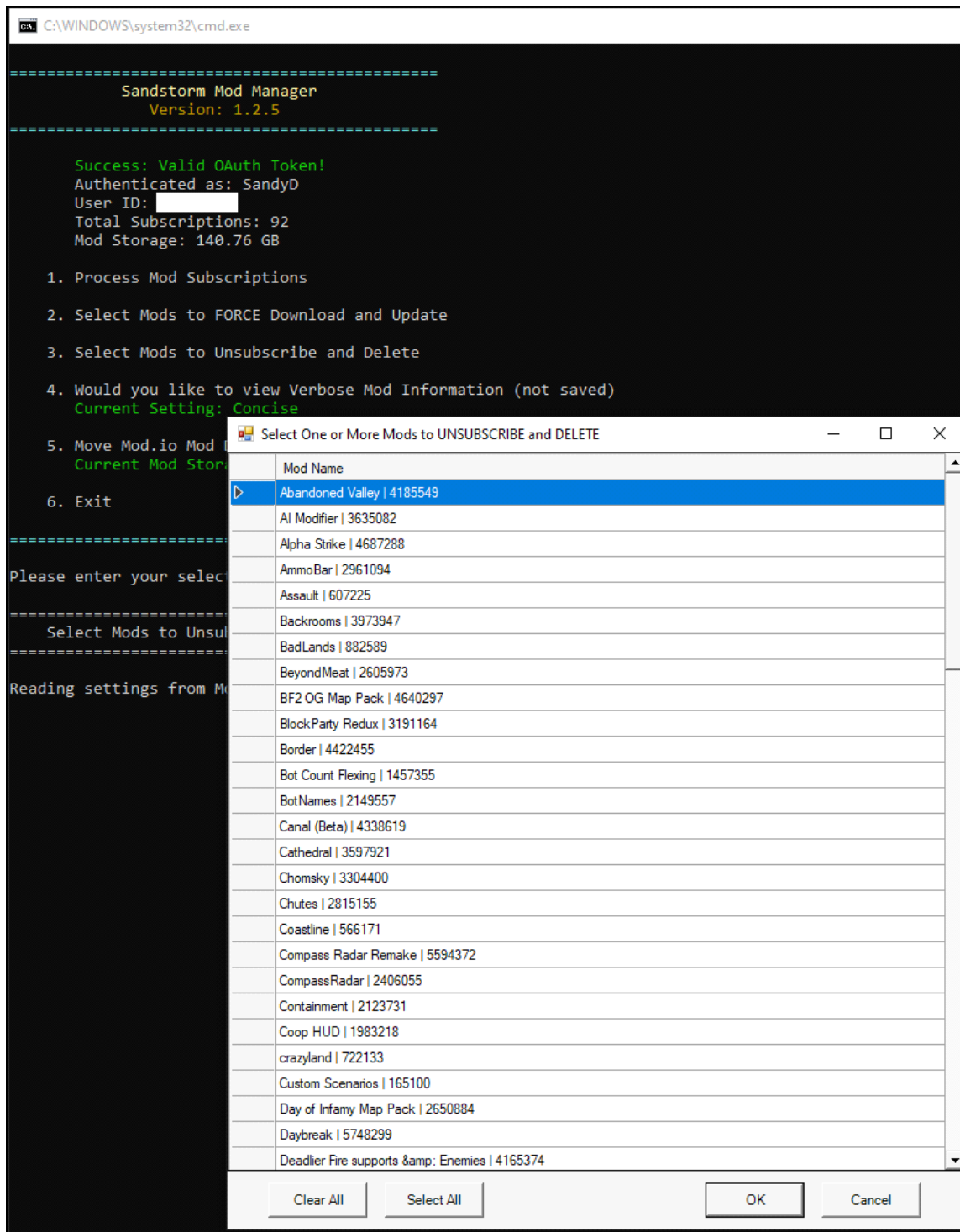
Reading settings from config.json.
You selected the following items:
138788 | Panzerschlacht
98145 | No Restricted Area Mutator [1.15]
98373 | More Ammo Mutator [1.15]

Last Chance: Do you really want to force an update for these items? [y/n]:
```

You will then be shown all of the mods you would like to re-download and ask if you really want to do it. If you enter Y(es) then the script will redownload those mods and check for any updates or new subscriptions.

3. Select Mods to Unsubscribe and Delete

```
=====
```



You will then be shown all of the mods you would like to Unsubscribe and ask if you really want to do it. You now have the option to Select All or Clear All.

```
=====
Select Mods to Unsubscribe and Delete
=====

Reading settings from ModList.json.

You selected the following items:
HitIndicator | 2435949
Hacienda | 5613035
Estates | 6096282

Last Chance: Do you really want to Unsubscribe and Delete these items? [y/n]:
```

If you enter Y(es) then the script will Unsubscribe to those mods and Delete the directories for those mods from your computer.

4. Would you like to view Verbose Mod Information

```
=====
```

Concise Display

Windows PowerShell

```
=====
Download your Subscriptions
=====
```

```
Found 1 Subscription Page.
```

```
Found 72 subscriptions.
```

```
=====
Processing 1 of 72 Subscriptions : Warehouse
```

```
Skipped Warehouse up to date
```

```
=====
Processing 2 of 72 Subscriptions : Estates
```

```
Skipped Estates up to date
```

```
=====
Processing 3 of 72 Subscriptions : Mashtuur
```

```
Skipped Mashtuur up to date
```

```
=====
Processing 4 of 72 Subscriptions : Daybreak
```

```
Skipped Daybreak up to date
```

```
=====
Processing 5 of 72 Subscriptions : Hacienda
```

```
Skipped Hacienda up to date
```

```
=====
Processing 6 of 72 Subscriptions : Compass Radar Remake
```

```
Skipped Compass Radar Remake up to date
```

```
=====
Processing 7 of 72 Subscriptions : The Open Pit
```

```
Skipped The Open Pit up to date
```

```
=====
Processing 8 of 72 Subscriptions : NoRestrictPlus
```

```
Skipped NoRestrictPlus up to date
```

Verbose display

```
Processing 70 of 72 Subscriptions : More Ammo Mutator [1.15]
modid: 98373
modFilename: filedata-5qlr.zip
filesize: 0.01 MB
moddate_added: 2023-10-30 19:46:51
modVersion: 0.8.4
modURL: https://g-254.modapi.io/v1/games/254/mods/98373/files/4468622/download

  Downloading More Ammo Mutator [1.15] - 98373 - filedata-5qlr.zip - 0.01 MB
  Unpacking More Ammo Mutator [1.15] - 98373 - filedata-5qlr.zip
7-Zip (a) 26.01 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-04-27
Scanning the drive for archives:
1 file, 12191 bytes (12 KiB)

Extracting archive: zip\filedata-5qlr.zip
--
Path = zip\filedata-5qlr.zip
Type = zip
Physical Size = 12191

Everything is Ok

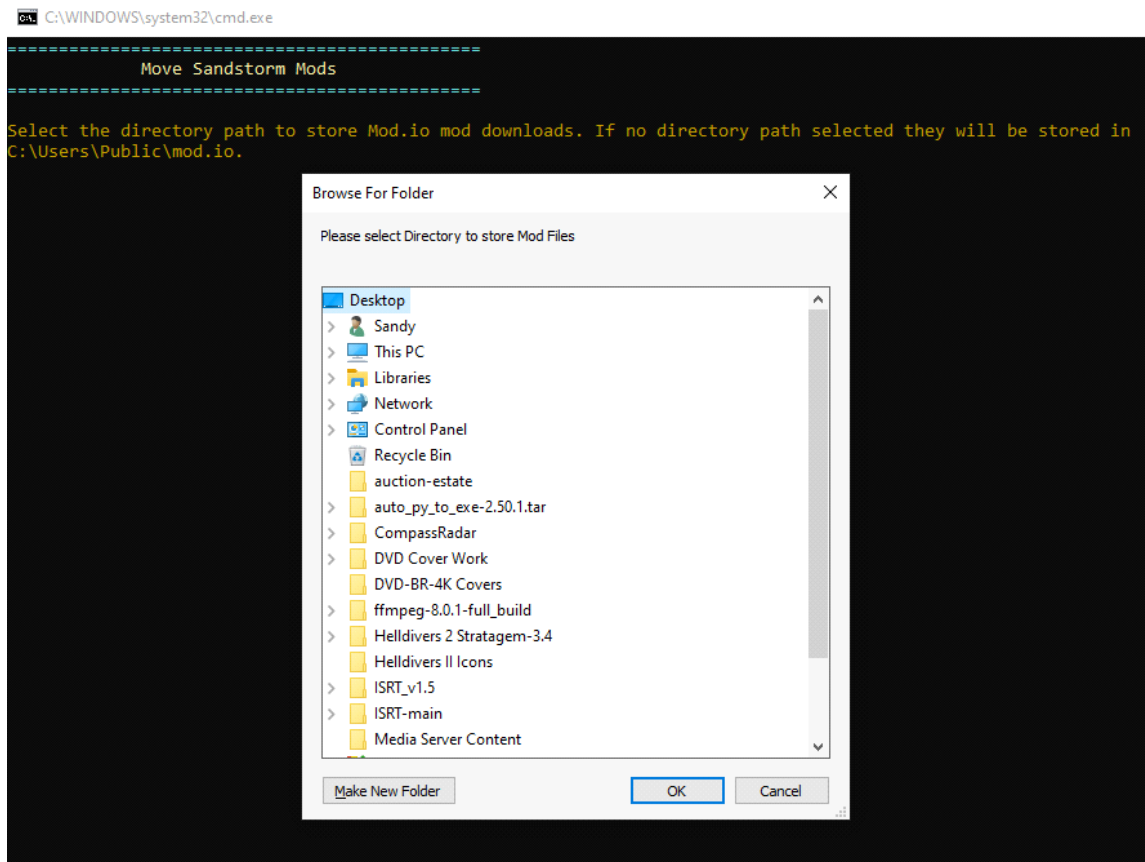
Size:          48477
Compressed: 12191
Success: Archive extracted with no errors.
=====
```

Toggle between Concise or Verbose display option when processing Subscriptions. (Not Saved)

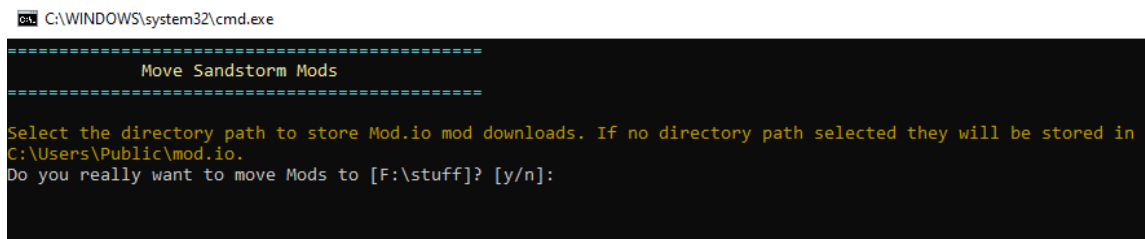
5. Move Mod.io Directory to new location

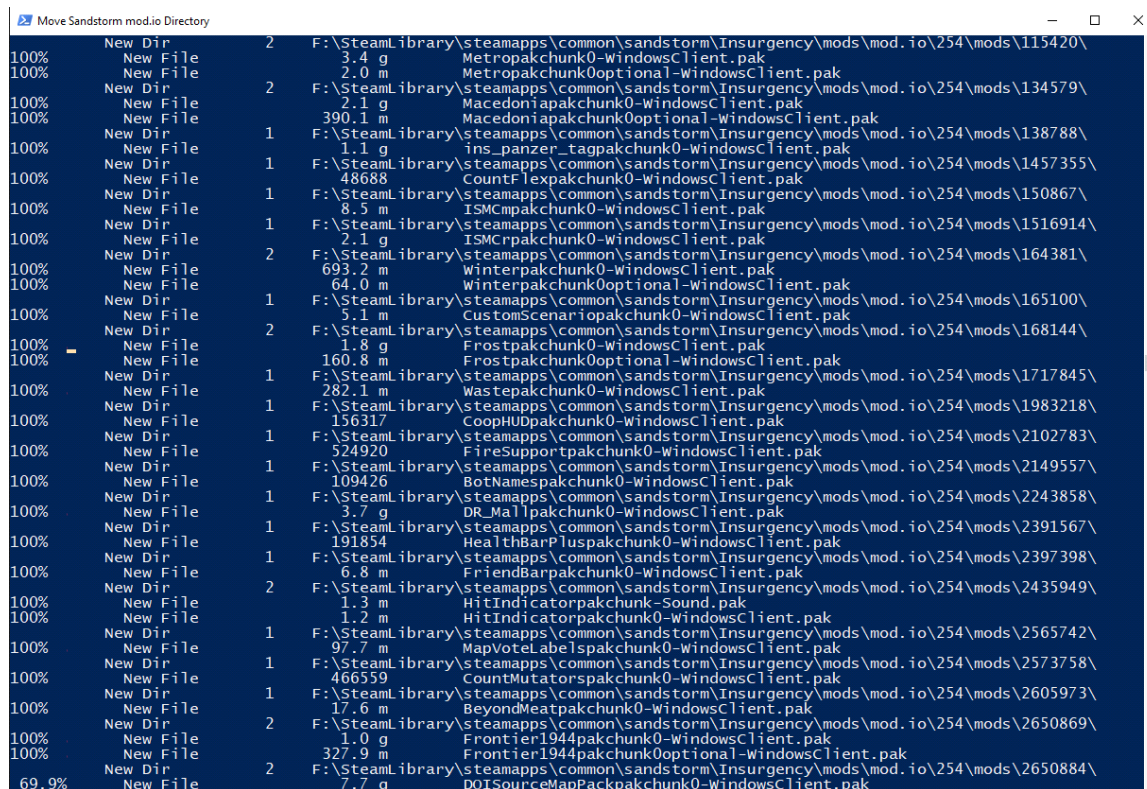
=====

You can move your Sandstorm Mod directory to any location on your computer. You no longer need to keep your mods on the C:\Users\Public\mod.io location. This will free up space on your C: drive. A GUI will open allowing you to select the location you would like the mods to be moved.

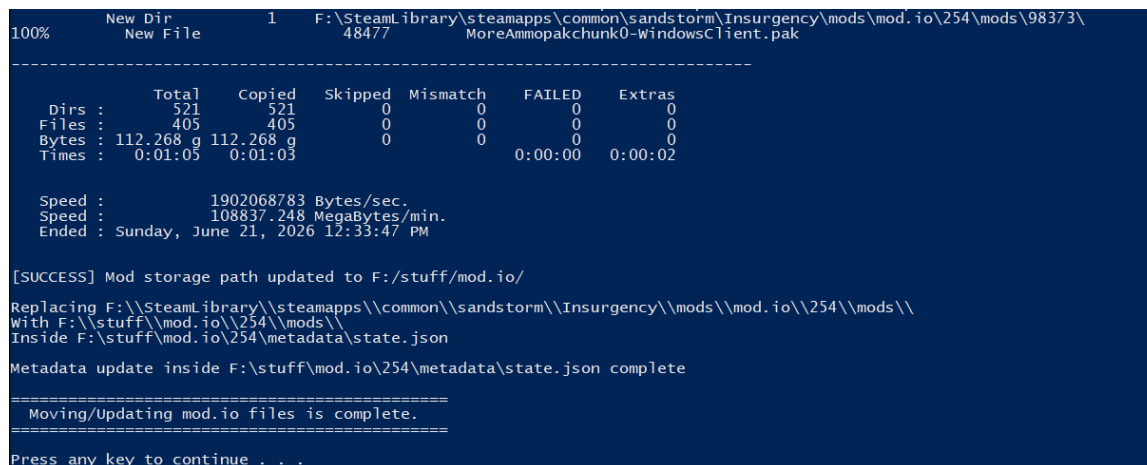


You will be given a chance to abort the move.





You will be shown the progress of moving every directory and file.



When finished you will be shown how many directories and files have been moved and if there were any errors.

6. Exit

=====

Kind of self explanatory. :)

Items of Note

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Server Operators

When you run this script to add newly subscribed mods to your server you may have to start the server twice. I have seen this happen when a server downloads new subscriptions on its own. It will sometimes stop booting while indicating the modlist is 0. When you restart the server it loads just fine. I have compared the state.json before and after with no change in it. I have no idea why it happens and it doesn't happen all of the time.

Players

If you join a server and it seems to hang on Downloading server mods... you can click the CANCEL button and then exit the game back to desktop. Then run this script as you have been subscribed to all of that server's mods and maps on mod.io. Running this script will download and install all of those mods to your game. When you load the game again it should go right into the server. You may get a short Downloading Server Mods... screen but just wait and you will get right in. After that you shouldn't see that screen again.

Final Note

The GUIs for Forced and Unsubscribed are now Alphabetized in ascending order

Change Log

=====

1.2.5 (2026-08-05)

1. [ADDED] On first execution script will scan state.json to build ModList.json. This will prevent downloading of all subscribed mods the first run.
2. [ADDED] "Select All" and "Clear All" buttons to the Force Download and Un-Subscribe GUIs. This way you can force a re-download of all mods or a mass unsubscribe of all mods without having to go to Mod.io
3. [ADDED] "Total Subscriptions: " to Main Menu
4. [ADDED] "Mod Storage: " showing total disk storage for all Mod downloads to Main
5. [ADDED] Directory path to where mods stored shown below Move Mod.io selection.

Menu1.2.1 (2026-07-25)

1. [FIXED] Missing \ in mod mover code placed the mods in the wrong directory.

1.2.0 (2026-07-23)

1. [CHANGED] The Sandstorm_mod_Mover.bat and directory has been removed. A much faster Powershell version has been added to Sandstorm_Mod_Manager.ps1 script.
2. [ADDED] Write-WrappedHost function to wrap long text lines to fit window width.
3. [ADDED] Modio_mod-directory function to open a GUI for select where to store mod files. If the GUI is canceled the selected directory will be C:\Users\Public\mod.io
4. [REMOVED] The Number 4 Testing feature
5. [ADDED] Only the last 11 state[DATE].json backup files are kept.
6. [DISABLED] globalsettings.backup restore as it was restoring on a false positive check.
7. [CHANGED] from Grid-Output GUI for Force DL and Unsubscribe to using System.Windows.Forms. This allows for more control over the window for style, positioning and size based upon the screen size.

1.1.4 (2026-07-18)

1. [CHANGED] When window opens it will now center horizontally and move to the top of the screen and extend to the task bar.

1.1.3 (2026-07-08)

1. [FIXED] When selecting test directory and it is drive root then there would be two // instead of just one /

1.1.2 (2026-07-08)

1. [FIXED] Sandstorm removes all spaces used in the User Account Name when creating the Directory inside %localappdata%\mod.io\254\[UserName] causing Sandstorm Mod Manager to not find the Directory. If the Windows Account you are using is My Name Sandstorm would shorten it to MyName and the Mod Manager wouldn't find it.

1.1.1 (2026-07-04)

1. [FIXED] When using the Mod Mover the mod location was not updated in the Mod Manager. If you tried processing, force downloading or unsubscribing to mods it would use the OLD location for storing and deleting mods instead of where they were moved.

1.1.0 (2026-07-02)

1. Listing total Subscription Count right after Page Count was not showing total Subscriptions but only the Subscription Count to the current page being processed.
2. Removed the need to create a Personal Access Token. The script uses the Games OAuth Token

stored in the user.json

3. Added backup and restore of globalsettings.json

Sometimes the game will reset the globalsettings.json to point back to C:\Users\Public\mod.io The script will make a globalsetting.backup file if one is not present. After that the script will compare the contents of the globalsettings.json with globalsettings.backup. If they do not match the contents of the backup will be restored to the json file. When you move the location of the mod.io directory the backup file will also be updated with the new location.

1.0.2 (2026-06-23)

1. When Processing Subscriptions is complete the total time downloading will be displayed in minutes:seconds, Total GB downloaded and Total GB of disk storage space used.

2. Added Start_Sandstorm_Mod_Manager.bat Double-Click on this file to start Sandstorm Mod Manager.

1.0.1 (2026-06-20)

1. Script will now make sure your Personal Access Token is valid every time you run the script and indicate it is valid in the menus and under who User Name and ID it belongs. If it detects an invalid PAT you will be taken to the Change Token menu item so you can enter a new token.

1.0 (2026-06-20) Initial Release

1. Token variable moved to a token.cfg file so you will not need to get a new token if you delete the config.json and haven't saved the old token id.

2. Added getting the mod directory from the %localappdata%\mod.io\globalsettings.json

3. Added a GUI selector for location if you want the mod files unpacked to a different location for testing.

4. A directory will now be created in the mod.io/mods directory using the mod_id to download and unpack the mod files.

5. config.json now contains the mod.io mod_id number and last update time for each mod for faster lookup of the latest update.

6. Mod.io json files have changed since the BoneLabModDownloader was created and it has been updated to work with the current system.

7. Dropped the internal Powershell Extractor for Zip files as it was limited and would fail if the file was larger than 4GB or if there were file crc errors that didn't corrupt the archive.

8. Added error reporting and a lot more information as mod files being downloaded and extracted.

9. If the numeric directory a mod is stored (ex: directory 123456 located in mod.io/254/mods/) has been deleted it will trigger an automatic download and update when script has been executed.
10. Completely rebuilding the state.json file using the subscribed mod information from mod.io.
11. Added Paging support for subscriptions that contain more than 100 mod files.
12. Added GUI to select mods for FORCED updating.
13. state.json is renamed to state_<date_time>.json before the new state.json is saved. This will give you backups incase the newly created state.json does not for some reason.
14. Auto delete any mod Unsubscribed on Mod.io
15. Added Elaped Download Time and Download Speed for each mod downloaded.
16. Added ability to unsubscribed to mods and delete the stored files.
17. Added Menu to more easily select what to do instead of a ordered process.
18. Added my Insurgency-Sandstorm-mod.io Mover batch file to the package and created a menu selection
19. Renamed config.json to ModList.json to better reflect contents
20. Both Forced Download and Unsubscribe GUI are now sorted alphabetically

To Do

=====

1. Add displaying of all Sandstorm Collections on Mod.io
2. Add displaying contents of a selected Sandstorm Collection
3. Add Follwing and Unfollowing Collections